



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Information Technology

Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: I Semester B.Tech. IT

Course Code & Title: GE3151 Problem solving and python programming

Name of the Faculty member (s): Mrs.G.Mareeswari

Innovative Practice Description

Unit / Topic: Unit III / Linear Search, Binary Search

Course Outcome: CO 3

Topic Learning Outcome: TLO 10

Activity Chosen: Think Pair Share

Justification:

- Students are asked to think individually and to share the knowledge with classmates. In python, Searching is an important concept and it is necessary for students to understand the searching concept and its execution. The students' individual understanding capability can be tested and the gained ideas can be shared by think-pair-share activity.
- Time Allotted for the Activity:** 15 minutes

Details of the Implementation:

- Instructor explained the particular concepts/topic in classroom within 35 minutes. Instructor gave a program to the students.
- Students think about what they know or have learned in searching, and come up with their own idea. [Takes 1-5 Minutes].
- The instructor asks them to pair with nearby students to discuss the program solution. The students interact with each other about their programs to deepen understanding of the iterative.
- They share their thoughts with each other and proceed with the task. [Takes 6-10 Minutes].
- Students share their solution with the entire class [Takes 11-15 Minutes]. Instructor moderates the discussion and highlights important points.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO 3	2	1	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
	2	1	1	1	1	1
Justification for correlation	It includes logical thinking and make the student able to think about problem solving approach	Able to approach the program by applying what they learnt	Able to provide solution for the searching problem	Students involved in the activity as team, the team work Skill improved	Students communication skill will be improved as they are facing audience and communicating the activity	Able to apply the problem solving approach to solve the complex searching problem.

• **Images / Screenshot of the practice:**



Fig:1 Glimpse of Think Pair Share Activity

(Ms. Harini. M share thought about the searching concept with her class mates,
Mr. Sanjay kumar write program for binary search.)

• **Reflective Critique:**

❖ ***Feedback of practice from students and other stakeholders:***

The students felt that this activity helps them to remember the concepts and it will be very helpful to correct their mistakes while sharing the program solution to the other students.

Benefit of the practice:

- From this activity, the students can get more clarity in the particular topic.
- Students can able to explain the concepts without any confusion.
- This practice has made the class more interactive than the conventional practice

Challenges faced in implementation:

- All the students were unable to complete the task within allotted time
- Not able to make all the students to involve in this activity
- Few students found it difficult to solve the program individually. After discussion, they felt easy.

References:

1. <https://www.ritrjpm.ac.in/images/computer-science/TPS-Venkat-GE6075.pdf>
2. <https://www.readingrockets.org/strategies/think-pair-share>
3. <https://www.wgu.edu/heyteach/article/how-think-pair-share-activity-can-improve-your-classroom-discussions1704.html>

Signature of Faculty Member

HOD